# Tony Wu

Massachusetts Institute of Technology | Mathematics and Computer Science

## Education

**Massachusetts Institute of Technology** Cambridge, MA
*B.S. in Mathematics and Computer Science* *Sep 2023 - May 2027 (expected)*
Cumulative GPA: 4.72/5.0

## Research Interests

Theoretical computer science, statistical machine learning, and AI safety, with research experience in graphical models, sigmoid-circuit lower bounds, and mechanistic estimation for neural networks.

## Publications

Learning Gaussian Graphical Models from a Glauber Trajectory Without Mixing.
Eric Shen*, **Tony Wu***, Mahbod Majid, and Ankur Moitra. **Equal first authors.*
*International Conference on Machine Learning (ICML), 2026.* Poster. arXiv:2606.31230

## Manuscripts under Review

Improved Rates of Learning Gaussian Graphical Models from Glauber Dynamics.
**Tony Wu**, Eric Shen, and Mahbod Majid.
*Submitted to ICLR 2027.*

## Research Experience

**Massachusetts Institute of Technology** Cambridge, MA
*Researcher, Circuit Complexity* *Sep 2025 - Present*
Advisor: Daniel Reichmann

- Study lower bounds for sigmoid circuits in theoretical computer science.
- Design circuit constructions for Boolean functions to investigate their computational complexity.

**Alignment Research Center** Berkeley, CA
*Visiting Researcher* *Jun - Aug 2026*

- Theoretically classified when mechanistic output estimation is possible or impossible depending on how adversarial the model is, focusing on multilayer perceptrons with shifted weights.
- Developed estimation algorithms and evaluated their performance empirically against baseline methods.

**Massachusetts Institute of Technology** Cambridge, MA
*Summer Program in Undergraduate Research (SPUR)* *Jun - Aug 2025*
Advisor: Ankur Moitra

- Co-developed a polynomial-time algorithm to learn the graph of a sparse Gaussian graphical model from a single Glauber trajectory, with a trajectory-length guarantee independent of mixing time (ICML 2026).
- Served as primary internal technical reviewer for the follow-up submitted to ICLR 2027, assessing its improved theoretical guarantees.

## Honors and Awards

**International Mathematical Olympiad (IMO)** Jul 2022
Gold Medal

## Teaching and Outreach

**Mathink** New South Wales, Australia
*Main Course Designer and Teacher* *May 2023 - Present*

- Design curricula that introduce students to university-level mathematics.
- Teach remotely and develop course materials for advanced mathematics instruction.

**Global Teaching Lab** Kigali, Rwanda
*Instructor* *Jan 2024*

- Taught Olympiad mathematics to 40 local teachers and several hundred students.
- Promoted the Rwandan mathematics program at six local high schools to encourage participation in mathematics competitions.

## Research Training

**FABRIC** Somerset, UK
*Program on AI and Reasoning (PAIR), Participant* *Jul - Aug 2024*

- Studied convolutional neural networks, transformers, and diffusion models.
- Explored mechanistic interpretability, steering vectors, and defenses against prompt injection.

## Selected Coursework

**Graduate mathematics.** Commutative Algebra; Algebraic Topology/Geometry; Number Theory.

**Graduate statistics and machine learning.** Statistical Inference; Information Theory; Non-asymptotic Statistics; Algorithms for Inference; Deep Learning.

**Undergraduate mathematics and computer science.** Algebra; Analysis; Topology; Differential Geometry; Computation Structures; Introduction to Programming.

## Technical Skills

Python, C/C++, LaTeX, Git, web design.